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Q1 (3 marks): We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 180 contain cats and 20 do not contain cats. The identification result of the model is that 140 images contain cats and 60 do not contain cats. Given that TP (true positive) of the model = 135. What is the Recall of the model?

	Actual	True	False	Total
Actual	135	45	180	
True	5	15	20	
False	140	60	200	

$$\text{Recall} = \frac{TP}{P} = \frac{135}{180} = 75\%$$

Q2 (2 marks): Using the following data set: (10, 25), (20, 70), (30, 380), (40, 550), (50, 610), (60, 1220), (70, 830), (80, 1450) in least squares method. Calculate the error when $x=30$, if you know that the linear regression equation is: $Y=19x-234$

x	y	h ₀	h ₁	h ₂	h ₃	h ₄	h ₅	h ₆	h ₇	h ₈	h ₉	h ₁₀	h ₁₁	h ₁₂	h ₁₃	h ₁₄	h ₁₅	h ₁₆	h ₁₇	h ₁₈	h ₁₉	h ₂₀	h ₂₁	h ₂₂	h ₂₃	h ₂₄	h ₂₅	h ₂₆	h ₂₇	h ₂₈	h ₂₉	h ₃₀	h ₃₁	h ₃₂	h ₃₃	h ₃₄	h ₃₅	h ₃₆	h ₃₇	h ₃₈	h ₃₉	h ₄₀	h ₄₁	h ₄₂	h ₄₃	h ₄₄	h ₄₅	h ₄₆	h ₄₇	h ₄₈	h ₄₉	h ₅₀	h ₅₁	h ₅₂	h ₅₃	h ₅₄	h ₅₅	h ₅₆	h ₅₇	h ₅₈	h ₅₉	h ₆₀	h ₆₁	h ₆₂	h ₆₃	h ₆₄	h ₆₅	h ₆₆	h ₆₇	h ₆₈	h ₆₉	h ₇₀	h ₇₁	h ₇₂	h ₇₃	h ₇₄	h ₇₅	h ₇₆	h ₇₇	h ₇₈	h ₇₉	h ₈₀	h ₈₁	h ₈₂	h ₈₃	h ₈₄	h ₈₅	h ₈₆	h ₈₇	h ₈₈	h ₈₉	h ₉₀	h ₉₁	h ₉₂	h ₉₃	h ₉₄	h ₉₅	h ₉₆	h ₉₇	h ₉₈	h ₉₉	h ₁₀₀	h ₁₀₁	h ₁₀₂	h ₁₀₃	h ₁₀₄	h ₁₀₅	h ₁₀₆	h ₁₀₇	h ₁₀₈	h ₁₀₉	h ₁₁₀	h ₁₁₁	h ₁₁₂	h ₁₁₃	h ₁₁₄	h ₁₁₅	h ₁₁₆	h ₁₁₇	h ₁₁₈	h ₁₁₉	h ₁₂₀	h ₁₂₁	h ₁₂₂	h ₁₂₃	h ₁₂₄	h ₁₂₅	h ₁₂₆	h ₁₂₇	h ₁₂₈	h ₁₂₉	h ₁₃₀	h ₁₃₁	h ₁₃₂	h ₁₃₃	h ₁₃₄	h ₁₃₅	h ₁₃₆	h ₁₃₇	h ₁₃₈	h ₁₃₉	h ₁₄₀	h ₁₄₁	h ₁₄₂	h ₁₄₃	h ₁₄₄	h ₁₄₅	h ₁₄₆	h ₁₄₇	h ₁₄₈	h ₁₄₉	h ₁₅₀	h ₁₅₁	h ₁₅₂	h ₁₅₃	h ₁₅₄	h ₁₅₅	h ₁₅₆	h ₁₅₇	h ₁₅₈	h ₁₅₉	h ₁₆₀	h ₁₆₁	h ₁₆₂	h ₁₆₃	h ₁₆₄	h ₁₆₅	h ₁₆₆	h ₁₆₇	h ₁₆₈	h ₁₆₉	h ₁₇₀	h ₁₇₁	h ₁₇₂	h ₁₇₃	h ₁₇₄	h ₁₇₅	h ₁₇₆	h ₁₇₇	h ₁₇₈	h ₁₇₉	h ₁₈₀	h ₁₈₁	h ₁₈₂	h ₁₈₃	h ₁₈₄	h ₁₈₅	h ₁₈₆	h ₁₈₇	h ₁₈₈	h ₁₈₉	h ₁₉₀	h ₁₉₁	h ₁₉₂	h ₁₉₃	h ₁₉₄	h ₁₉₅	h ₁₉₆	h ₁₉₇	h ₁₉₈	h ₁₉₉	h ₂₀₀
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$$e = y_d - y_{\text{predicted}}$$

$$e = 380 - 336 = 44$$

Q3 (3 marks): For each of the following application mention the best learning method (Supervised, Unsupervised, or reinforcement):

- Robot navigation ~~supervised~~ reinforcement
- Customer segmentation unsupervised
- Character recognition supervised

Q4 (3 marks): Mention three methods can be used to speed up Back-propagation learning process.

1) considering β

Q5 (5 marks): For the following assume: inputs range from 0 to 255, output 1 or 0, linear separable, learning rate 0.1, initialize all weights to 0.5.

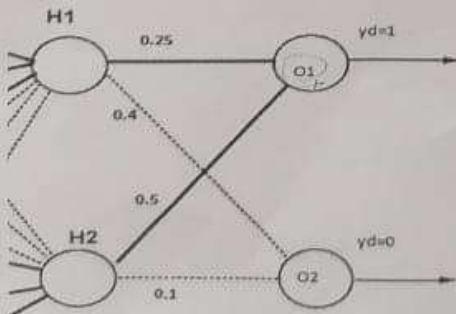
- A- Draw the needed ANN to recognize the following pattern [20, 113, 9, 230]
- B- Calculate the ANN output for the following inputs:

$$[0, 0, 2, 50] \quad y_1 = (0.5 * 0 + 0.5 * 0 + 0.5 * 2 + 0.5 * 50) - 0.1 = 25.9$$

Q6 (6 marks):

Given a multi-layer network with two output nodes (shown Figure) trained with backpropagation algorithm, assume the activation functions used are the sigmoid functions.

Assume the output from H_1 & H_2 was 0.9 and 0.1 respectively. Answer the following:



$$\delta_k(p) = y_k(p) \times [1 - y_k(p)] \times e_k(p),$$

$$\delta_j(p) = y_j(p) \times [1 - y_j(p)] \times \sum_{k=1}^i \delta_k(p) w_{jk}(p),$$

$$b) \delta_1 = 0.5187 (1 - 0.5187) \times 0.4813 = 0.1201$$

$$e_1 = 1 - 0.5187 = 0.4813$$

$$c) \Delta w = 0.9 * 0.1 * 0.1201 = 0.010809$$

$$\theta = 0,2$$

- a) Calculate the output of O_1 ? $y_{o_1} = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(0,9 \times 0,25 + 0,1 \times 0,2)}} = 0,5187$
- b) Calculate the error gradient for O_1 ?
- c) Calculate change for the weight (Δw) between H_1 and O_1 assume a learning rate of 0.1?

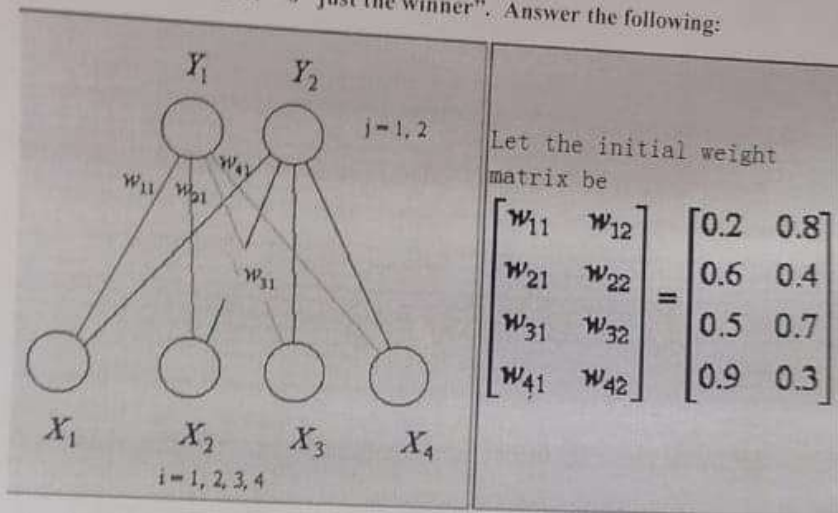
Q7 (8 marks): Multiple choice questions:

A. Consider a Kohonen self-organizing map in the Figure below. There are only four input training patterns:

1 1 0 0, 0 0 0 1, 1 0 0 0, 0 0 1 1

Suppose $\alpha = 0.6$

Let neighbor radius $R = 0$ "just the winner". Answer the following:



- For the first vector presented to the network 1 1 0 0, D1 (Distance to first cluster Y_1) =
 - 2.276
 - 66
 - 1.86
 - 1.36
 - none
- D2 (Distance to second cluster Y_2) for first vector:
 - 0.6
 - 0.98
 - 1
 - 0.68
 - none
- Updating weights after presenting first vector:

a.	0.68	0.8	b.	0.08	0.92
	0.84	0.4		0.24	0.76
	0.8	0.7		0.2	0.28
	0.96	0.3		0.96	0.12
c.	0.2	0.92	d.	0.68	0.8
	0.6	0.76		0.84	0.4
	0.5	0.28		0.8	0.28
	0.9	0.12		0.96	0.12

e. None
- Which type of learning does a SOM use?
 - Supervised
 - Unsupervised
 - Reinforcement
 - Semi-supervised

Q1 (4 points): What is the difference between overfitting and underfitting?

overfitting :-

high bias low variance

underfitting :-

low bias high variance

regression :-

continuous output

classification :-

Q3 (6 points): Multiple choice

A- In simple linear regression, there is _____ dependent variable and _____ independent variable(s)

1. One, multiple
2. Multiple, one
3. One, one
4. Multiple, multiple

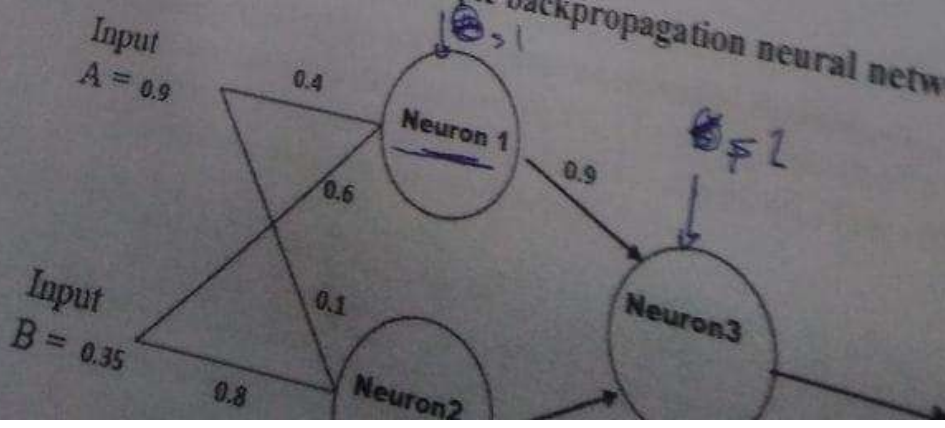
B- Playing a chess game machine learning model is a _____ learning:

- a. supervised
- b. unsupervised
- c. reinforcement
- d. neural network

C- Function approximation

- a. supervised
- b. unsupervised
- c. unknown
- d. neural network

Q4 (10 marks): Consider the simple backpropagation neural network below:



overfitting:

under fitting: Large training error

over fitting

Q2 (2 points): What is the difference between regression and classification?

regression: يعطى بالكميات

classification: يعطى قية (قيمة)

(2)

Q3 (6 points): Multiple choice

A- In simple linear regression, there is _____ dependent variable and _____ independent variable(s)

1. One, multiple
2. Multiple, one
3. One, one
4. Multiple, multiple

ع.ب.ب.ب.

B- Playing a chess game machine learning model is a _____ learning:

- a. supervised b. unsupervised c. reinforcement d. neural network

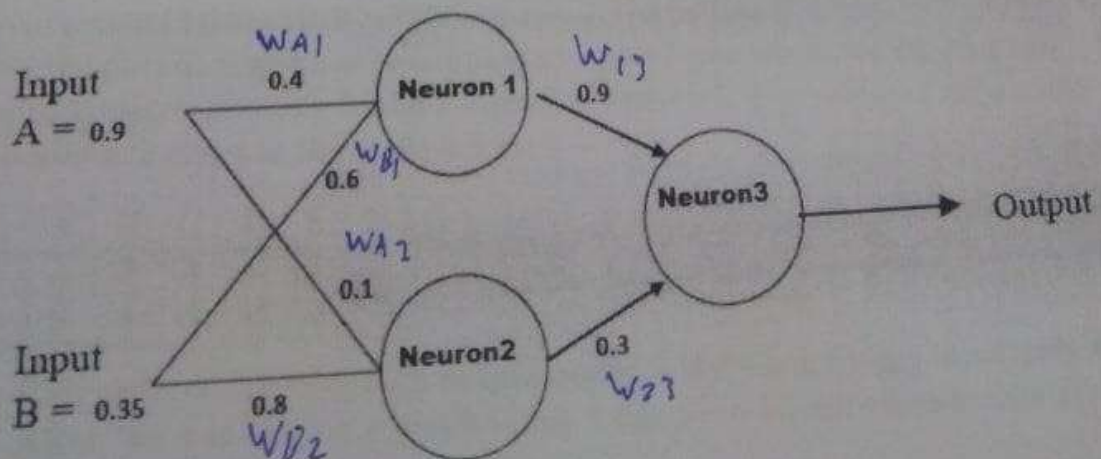
ع.ب.ب.ب.

C- Function approximation _____ learning:

- a. supervised b. unsupervised c. unknown d. neural network

ع.ب.ب.ب.

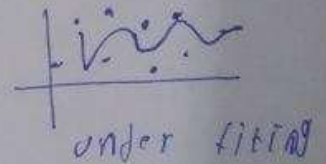
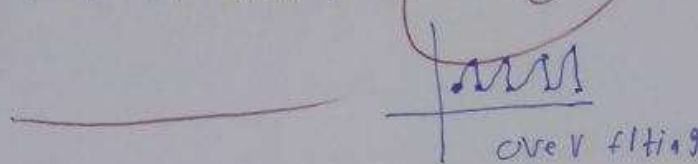
Q4 (10 marks): Consider the simple backpropagation neural network below:



Q1 (4 points): What is the difference between overfitting and underfitting? Suggest a method to avoid overfitting and another to avoid underfitting.

overfitting: Small training error, testing error Large

under fitting: Large training error



Q2 (2 points): What is the difference between regression and classification?

regression: يعطيه بالكميات

classification: يعطيه قيمه (رقمية) (2)

Q3 (6 points): Multiple choice

A- In simple linear regression, there is _____ dependent variable and _____ independent variable(s)

1. One, multiple
- ☒ 2. Multiple, one
3. One, one
4. Multiple, multiple

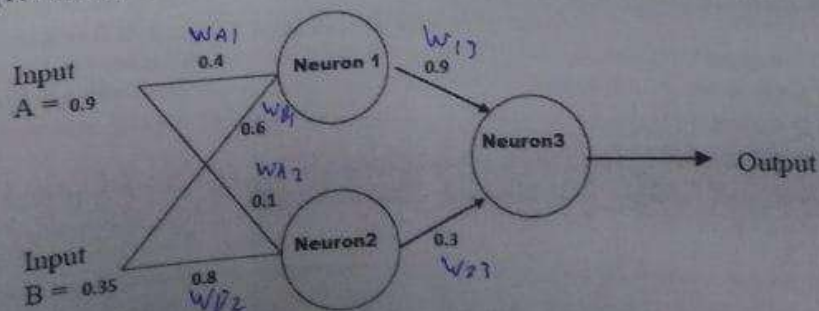
B- Playing a chess game machine learning model is a _____ learning:

- a. supervised
- b. unsupervised
- c. reinforcement
- ☒ d. neural network

C- Function approximation learning:

- a. supervised
- b. unsupervised
- ☒ c. unknown
- d. neural network

Q4 (10 marks): Consider the simple backpropagation neural network below:



Assume threshold $\theta=0$ and learning rate=1 and target (desired output) = 0.5. Compute the following:

- Output of neuron 1:
- Output of neuron 2:
- Output neuron 3 gradient error δ :
- W_{13} new?
- Gradient error for hidden layer δ_{1j} :

$$\delta_k(p) = y_k(p) \times [1 - y_k(p)] \times e_k(p)$$

$$\delta_j(p) = y_j(p) \times [1 - y_j(p)] \times \sum_{k=1}^l \delta_k(p) w_{jk}(p)$$

$$a) Y_1 = \text{sigmoid}(W_{A1} A + W_{B1} B) - \theta = \text{sigmoid}(0.4 \times 0.1 + 0.6 \times 0.15 - 0) = \frac{1}{1+e^{-0.1}} = 0.16$$

$$b) Y_2 = \text{sigmoid}(W_{A2} A + W_{B2} B) - \theta = \text{sigmoid}(0.1 \times 0.1 + 0.8 \times 0.15 - 0) = \frac{1}{1+e^{-0.13}} = 0.15$$

$$c) Y_3 = \text{sigmoid}(W_{13} Y_1 + W_{23} Y_2) - \theta = \text{sigmoid}(0.9 \times 0.16 + 0.1 \times 0.15 - 0) = \frac{1}{1+e^{-0.151}} = 0.16$$

$$e = Y_3 - Y_{\text{act}} = 0.15 - 0.16 = -0.01$$

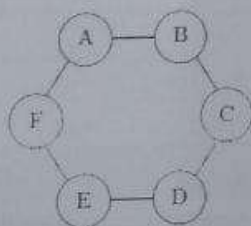
$$\delta_3 = Y_3 (1 - Y_3) e = 0.16 (1 - 0.16) \times -0.01 = -0.0024$$

$$d) W_{13} \text{ new} = W_{13} \text{ old} + \Delta W = 0.9 + 1 \times Y_1 \times \delta_3 = 0.9 + 1 \times 0.16 \times -0.0024 = 0.88$$

$$e) \delta_{13} = Y_1 (1 - Y_1) \sum W_{13} \delta_3$$

$$= 0.16 (1 - 0.16) \times 0.9 \times -0.0024 = -5.18 \times 10^{-5}$$

Q5 (8 marks) SOM. Below is a diagram of a self-organizing map:



The output layer of this map consists of six nodes, A, B, C, D, E and F, which are organized into a two dimensional lattice with neighbors connected by lines.

Each of the output nodes has two inputs x_1 and x_2 (not shown on the diagram). Thus, each node has two weights corresponding to these inputs: w_1 and w_2 . The values of the weights for all output in the SOM nodes are given in the table below:

Node	A	B	C	D	E	F
w_1	-1	0	3	-2	3	4
w_2	2	4	-2	-3	2	-1

- Calculate which of the six output nodes is the winner if the input pattern is $x = (2, -4)$?
- After the winner for a given input x has been identified, calculate the winner new weights. Let $\alpha = 0.5$.

$$a) A = -1 \times 2 + 2 \times -4 = -10$$

$$B = 0 \times 2 + 4 \times -4 = -16$$

$$D = -2 \times 2 + -3 \times -4 = -16$$

$$E = 3 \times 2 + 2 \times -4 = -2$$